<u>ASSIGNMENT 12:</u> Deploy and run the project in AWS without using the port.

Note: Login to your AWS Account. And do as mentioned in Assignment 10.

Step 1: First Go to root using '***cd ~*** 'and then change the directory using '***cd /etc/nginx/sites-available***'. After that run '***sudo nano default***' command as follows.

```
ubuntu@13.233.126.36:22 - Bitvise xterm - ubuntu@ip-172-31-11-121: ~/DS
ubuntu@ip-172-31-11-121:~/DS$ cd ~
ubuntu@ip-172-31-11-121:~$ cd /etc/nginx/sites-available
ubuntu@ip-172-31-11-121:/etc/nginx/sites-available$ sudo nano default
```

Step 2: The following file opens up in nano editor.

```
ubuntu@13.233.126.36:22 - Bitvise xterm - ubuntu@ip-172-31-11-121: /etc/nginx/sites-available
  GNU nano 7.2                              default
##
# You should look at the following URL's in order to grasp a solid understanding
# of Nginx configuration files in order to fully unleash the power of Nginx.
# https://www.nginx.com/resources/wiki/start/
# https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/
# https://wiki.debian.org/Nginx/DirectoryStructure
#
# In most cases, administrators will remove this file from sites-enabled/ and
# leave it as reference inside of sites-available where it will continue to be
# updated by the nginx packaging team.
#
# This file will automatically load configuration files provided by other
# applications, such as Drupal or Wordpress. These applications will be made
# available underneath a path with that package name, such as /drupal8.
#
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
##

# Default server configuration
#
server {
        listen 80 default_server;
        listen [::]:80 default_server;

        # SSL configuration
        #
        # listen 443 ssl default_server;
        # listen [::]:443 ssl default_server;
        #
        # Note: You should disable gzip for SSL traffic.
        # See: https://bugs.debian.org/773332
        #
        # Read up on ssl_ciphers to ensure a secure configuration.
        # See: https://bugs.debian.org/765782
        #
        # Self signed certs generated by the ssl-cert package
        # Don't use them in a production server!
        #
        # include snippets/snakeoil.conf;

        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;

        server_name _;

        location / {
                # First attempt to serve request as file, then
                # as directory, then fall back to displaying a 404.
                try_files $uri $uri/ =404;
        }

        # pass PHP scripts to FastCGI server
        #
        #location ~ \.php$ {
        #       include snippets/fastcgi-php.conf;
        #
        #       # With php-fpm (or other unix sockets):
        #       fastcgi_pass unix:/run/php/php7.4-fpm.sock;
        #       # With php-cgi (or other tcp sockets):
        #       fastcgi_pass 127.0.0.1:9000;
        #}

        # deny access to .htaccess files, if Apache's document root
        # concurs with nginx's one
        #
        #location ~ /\.ht {
        #       deny all;
        #}
}


# Virtual Host configuration for example.com
#
# You can move that to a different file under sites-available/ and symlink that
# to sites-enabled/ to enable it.
#
#server {
#       listen 80;
#       listen [::]:80;
#
#       server_name example.com;
#
#       root /var/www/example.com;
#       index index.html;
#
#       location / {
#               try_files $uri $uri/ =404;
#       }
#}
^G Help       ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location   M-U Undo
^X Exit       ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

Step 3: Modify the location portion as given below. Then press '***ctrl+O***' and hit Enter which will save the modified portion. Then press '***ctrl+X***' to exit from the file editor.

```
ubuntu@13.233.126.36:22 - Bitvise xterm - ubuntu@ip-172-31-11-121: /etc/nginx/sites-available
  GNU nano 7.2                              default *
        # Self signed certs generated by the ssl-cert package
        # Don't use them in a production server!
        #
        # include snippets/snakeoil.conf;

        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;

        server_name _;

        location / {
                # First attempt to serve request as file, then
                # as directory, then fall back to displaying a 404.
                #try_files $uri $uri/ =404;
                proxy_pass http://localhost:4000;
                proxy_http_version 1.1;
                proxy_set_header Upgrade $http_upgrade;
                proxy_set_header Connection 'Upgrade';
                proxy_set_header Host $host;
                proxy_cache_bypass $http_upgrade;
        }

        # pass PHP scripts to FastCGI server
        #
        #location ~ \.php$ {
        #       include snippets/fastcgi-php.conf;
        #
        #       # With php-fpm (or other unix sockets):
        #       fastcgi_pass unix:/run/php/php7.4-fpm.sock;
        #       # With php-cgi (or other tcp sockets):
^G Help       ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location   M-U Undo
^X Exit       ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

Step 4: Now run the command '***sudo chmod 777 default*** ' to give all users full permission (read, write, execute) to the file and then '***sudo systemctl restart nginx*** ' to restart the nginx. After that change the directory to git cloning directory by '***cd ~*** ' and '***cd <directory name>/*** '. Then start the server using '***node index.js*** ' command.

```
ubuntu@13.233.126.36:22 - Bitvise xterm - ubuntu@ip-172-31-11-121: ~/DS
ubuntu@ip-172-31-11-121:/etc/nginx/sites-available$ sudo chmod 777 default
ubuntu@ip-172-31-11-121:/etc/nginx/sites-available$ sudo systemctl restart nginx
ubuntu@ip-172-31-11-121:/etc/nginx/sites-available$ cd ~
ubuntu@ip-172-31-11-121:~$ cd DS/
ubuntu@ip-172-31-11-121:~/DS$ node index.js
Started server
```

Step 6: Now go to a new web browsing page and paste the IP Address of your Instance. The web page(index.js) is running successfully without using any port (4000).